

EE 536: IoT Systems

Course Code	EE 536
Course Name	IoT Systems
L-T-P-C	2-0-2-3
Prerequisites	Communication theory (EE304); Computer networks (CS406); Microcontroller programming and Digital systems design (IC161)
Intended for	B.Tech. III/IV year/ MS/M.Tech./PhD
Distribution	Core for M.Tech. (CSP)
Approval	33rd BoA (Jan 2020); 24th Senate (Feb 2020)

Course Contents

- **An introduction to IoT systems:** [1 Lectures]
 - Introduction and motivation of IoT systems
- **Hardware components of IoT systems:** (2 Lectures + 2 lab hours)
 - A quick overview of different components---micro-controllers, SoCs, communication modules, power supply and sensing modules---of off-the-shelf prototyping boards, e.g., Arduino UNO, MSP430 LaunchPad; NodeMCU, STM32, Raspberry Pi.
- **Software component of IoT systems:** (6 Lectures + 6 lab hours)
 - Introduction to IDEs for off-the-shelf boards, e.g., Arduino IDE, Waspnote IDE, Code composed studio; Contiki-OS and RIOT OS; 6LowPAN network stack; Sensor interfacing; GPIO programming
- **Communication paradigm of IoT systems:** (12 Lectures + 12 lab hours)
 - Different wireless standards, e.g., IEEE802.15.4, ZigBee, BLE, IEEE802.11; link layer technologies, Medium Access Control; Routing; Application layer protocols; Network topologies.
- **Performance evaluation of IoT systems:** (4 Lectures + 4 lab hours)
 - Developing mathematical models for energy consumption, Optimal node placement, resource allocation over wireless sensor networks to meet QoS requirements.
- **Case studies/mini projects:** (3 Lectures + 4 lab hours)
 - Home automation; Building energy management; Indoor positioning; Air quality monitoring; Precision agriculture; Smart parking

Text Books

1. David Hanes et al., **IoT Fundamentals: Networking Technologies, Protocols and Use Cases for the Internet of Things**, Pearson, 2017

References

1. Parry Lea, **Internet of Things for Architects: Architecting IoT Solutions by Implementing Sensors, Communication Infrastructure**, Edge Computing, Analytics, and Security, Packt Publishing Limited, 2018
2. Shuang-Hua Yang, **Wireless Sensor Networks: Principles, Design and Applications**, Springer.
3. Kazem Sohraby, Daniel Minoli, Taieb F. Znati, **Wireless Sensor Networks: Technology, Protocols, and Applications**, Wiley Interscience, 2009
4. White papers, RFCs, survey articles on Wireless communication standards and technologies.
5. Antonio Linan Colina, Alvaro Vives, Antoine Bagula, Marco Zennaro and Ermanno Pietrosemoli, **IoT in five Days**, <https://github.com/marcozennaro/IPv6-WSN-book/releases/>